



ORBITAL MANEUVERS USING GRAVITATIONAL CAPTURE TIMES

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ABSTRACT

Artificial satellites around the Earth can be temporarily captured by the Moon via gravitational mechanisms. How long the capture remains depends on the phase space region where the trajectory is located. This interval of time (capture time) ranges from less than one day (a single passage), up to 500 days, or even more. Orbits of longer times might be very useful for certain types of missions. The advantage of the ballistic capture is to save fuel consumption in an orbit transference from around the Earth to around the Moon. Some of the impulse needed in the transference is saved by the use of the gravitational forces involved. However, the time needed for the transference is elongated from days to months. In the present work we have mapped a significant part of the phase space of the Earth-Moon system, determining the length of the capture times and the origin of the trajectory, if it comes from the Earth direction, or from the opposite direction. Using such map we present a set of missions considering the utilization of the long capture times.

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INTRODUCTION

The gravitational capture is a temporary phenomenon (see, for example, Hopf (1930)). However, many authors applied this mechanism as a technique to save fuel, such as Belbruno (1987, 1990, 1992), Krish (1991), Krish et al. (1992), Miller and Belbruno (1991), Belbruno and Miller (1990, 1993), Yamakawa et al. (1992, 1993). The first real application of a ballistic capture transfer was made during an emergency in a Japanese spacecraft (Belbruno and Miller, 1990).

In this paper we studied the behavior of the time that a spacecraft spend in the vicinity of the Moon for several initial conditions. In general, the gravitational capture is studied in the frame of the restricted three-body problem, through numerical simulations. The complexity of this problem can be seen in Cordeiro et al. (1999). A precise definition of gravitational capture is not well established. In the present work, a capture, or escape trajectory, will be defined for trajectories which has a sign change of its orbital (two body) energy. If the sign changes from positive to negative it is a capture, conversely it is an escape.

This work is also an extension of an early study of the capture time done by Vieira Neto and Prado (1998). In that work they studied the minimum energy, which leads to very eccentric trajectories ($e > 0.98$). The space a vs. e was explored by Winter and Vieira Neto (2002) instead of the usual perilune distance and it gives more trajectories possibilities (eccentricities from 0 to 0.99). To classify the capture times we used the same methodology applied in Vieira Neto and Winter (2001) but using the space R_p vs. E , the perilune distance and the two-body energy. We used a purely gravitational mechanism to explore the capture time for a certain range of initial conditions.

MATHEMATICAL MODEL AND JACOBI CONSTANT

The Earth-Moon-particle system will be studied using the planar circular restricted three-body problem. In this model Earth and Moon are moving in circular orbits about their common center of mass at a fixed angular velocity. The direction of the x axis is chosen such that the two masses always lie along it with the Moon on the positive side of the axis. This dynamical model is not integrable, and the total energy and angular momentum of the particle are not conserved. Nevertheless, there is one integral of motion, named Jacobi Constant (C_J) which can be used to find limits of forbidden regions, known as zero velocity curves. The values of the Jacobi constant at the Lagrangian points can be used to determine whether an initial condition can escape from the Moon, or not. For particles which have C_J values larger than the value at L_1 , the inner Lagrangian point, the orbits are constrained at three possible regions: around the Earth; exterior to the prohibited region; and at a small region around the Moon. In such case, a particle around the Moon can not cross the prohibited region to go around the Earth, neither escape to outside of the Earth-Moon system, i.e., it can not escape from its orbit around the Moon. Therefore, for trajectories which have $C_J \geq C_{JL1}$ it is impossible to occur a temporary capture by the Moon. However, it can never reach the Moon if it is outside the Moon's vicinity. Therefore, the Jacobi constant is used in order to select initial conditions whose orbits might perform a gravitational capture ($C_J < C_{JL1}$).

TEMPORARY GRAVITATIONAL CAPTURE

To characterize a gravitational capture the two-body Energy is monitored. Due to the perturbation of the third body (Earth), the two-body energy (Moon-particle) is not constant. The changing of the energy sign indicates that the osculating orbit changed from a closed orbit to an open one (escape) or from an open orbit to a closed one (capture).

It is necessary a non conservative process to turn a temporary gravitational capture into a permanent one. The natural dissipative processes are slow and need relatively long times to be effective, but spacecrafts can use impulsive forces to stabilize a capture around the Moon. Therefore, it is relevant to find gravitational capture trajectories which offer a minimum fuel consumption (low energy transfers) with the minimum time duration of the transfer mission. However, other situations are also possible and give interesting results. For example, if the spacecraft needs to stay in the vicinity of the Moon for long time period measuring some property, it is possible to use trajectories with long times to perform this mission and there will be no fuel consumption to maintain this trajectory around the Moon.

ANALYSIS OF THE ENERGY AND THE TIME OF FLIGHT

In this section we investigate the effect of the initial conditions of the osculating two-body energy and pericenter position around the Moon. This numerical study determines regions for the capture time distribution. We name "capture time" as the time that the particle remains with a negative energy around the Moon after we release it from its initial position. A gray-scale is used to show structures in terms of the capture times. The work is restricted into two possibilities, the pericenter longitude at $\varpi = 0^\circ$ and at $\varpi = 180^\circ$, that is, in opposition and in inferior conjunction, respectively, where the mirror theorem works. This choice was made due to the results of Vieira Neto and Prado (1998), and Yamakawa *et al.* (1992, 1993), which show that these two choices of the pericenter longitude give the most economic trajectories (the value of the two-body energy can reach the most negative values). All trajectories are initially prograde. The results are shown in Figs. 1 and 2.

In these figures we see the length of time in the vicinity of the Moon, the pericenter distance, and the two-body energy of the particle. The capture time have an exponential structure, and the longest capture times are on the border defined by the Jacobi constant at L_1 . The white island inside the grey region, is a region where the trajectories do not escape during the integration time of 5 000 days. Winter and Vieira Neto (2002) showed that such region are composed of quasi-periodic trajectories, and no matter how long the integration time is, those trajectories will never escape. The border of the white island also provides trajectories with long capture times.

Another regions present in the pictures are the collision regions, represented by the small plus sign. Since the initial conditions are actually the final conditions of the trajectory, the collisions occur when the

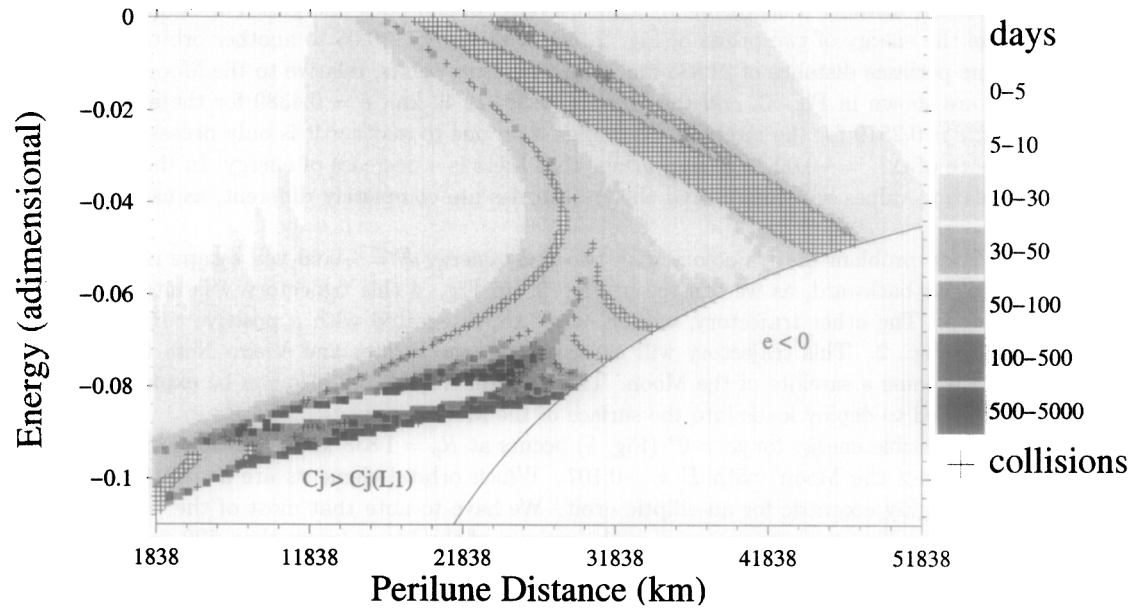


Fig. 1. Capture times in the initial condition space of energy and perilune distance. Capture time, in days, is represented by gray-scale. The range of initial conditions considered are delimited by the zero velocity curves associated to L_1 and eccentricities smaller than zero. The particles are initially at pericenter with $\varpi = 0^\circ$. The white region inside the range of initial conditions considered correspond to trajectories that surpass 5000 days without escape, called "prisoner trajectories" (Vieira Neto and Winter, 2001). Crosses correspond to trajectories which ends in collision with the Moon.

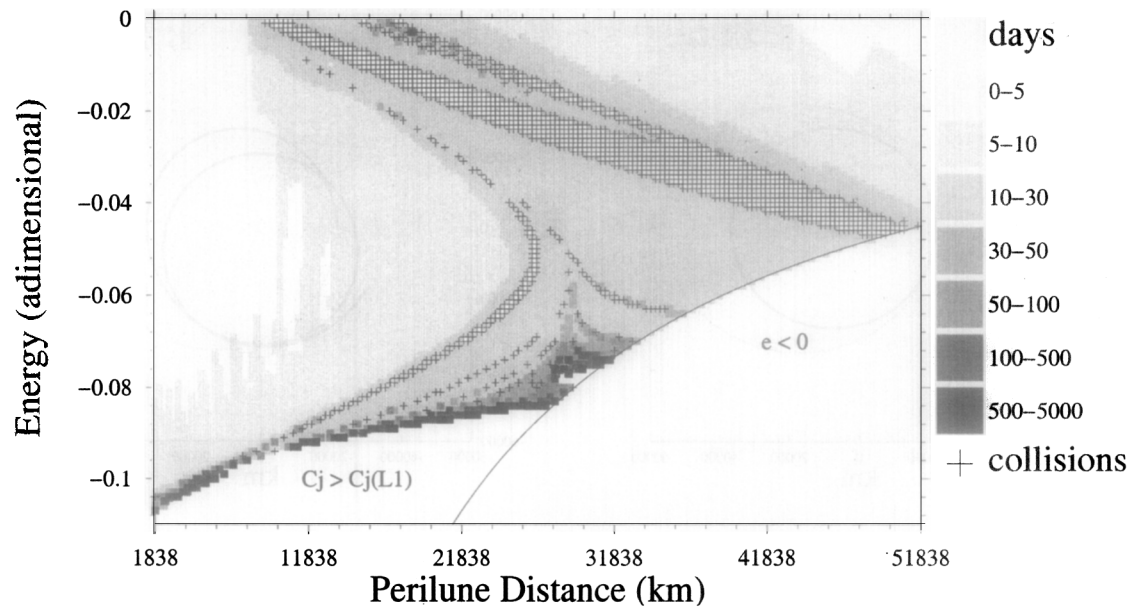


Fig. 2. Same as in Fig. 1, but for $\varpi = 180^\circ$.

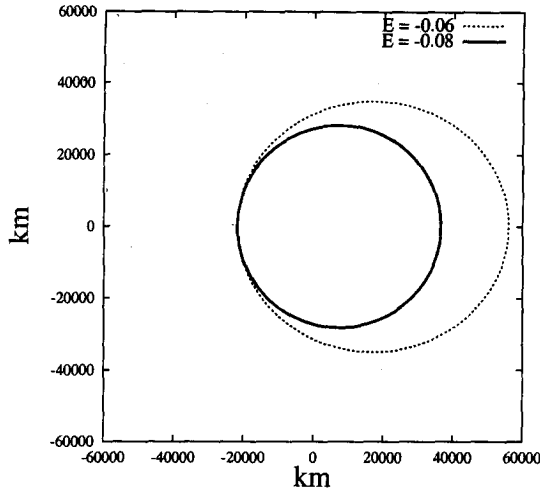


Fig. 3. Two osculating orbits around the Moon. They have energy $E = -0.06$ and $E = -0.08$, and both have perilune distance of $R_p = 21\,838$ km. The perilune longitude is $\varpi = 180^\circ$.

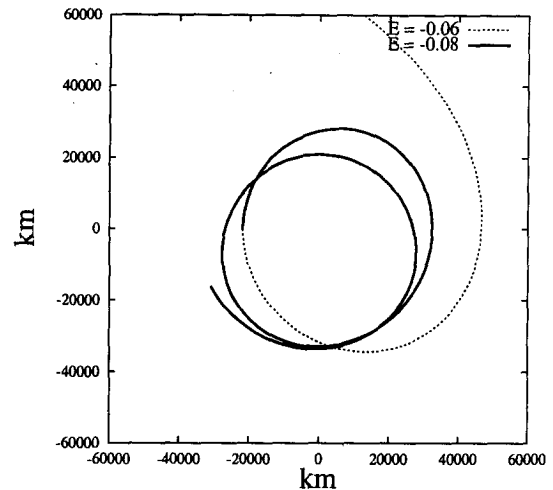


Fig. 4. The real trajectories of Fig. 3 for the three-body problem. The trajectory with energy $E = -0.06$ is an escape trajectory and was integrated with a negative time step. The other trajectory ($E = -0.08$) will not escape (see text) and was integrated with a positive time step.

particles still with a negative energy. So, the collision regions are also inaccessible regions by a spacecraft outside the Moon's gravitational sphere of influence, where the particle has a positive energy, relative to the Moon.

For a fixed perilune distance it is possible to achieve a range of two-body energies. For example, suppose we want to change the energy of two orbits of Fig. 2, one with energy -0.06 to another orbit with -0.08 , both with the same perilune distance of $21\,838$ km. The osculating orbits, relative to the Moon, with these initial conditions are shown in Fig. 3, and they have $a = 38\,922.42$ km, $e = 0.4389$ for the external, and $a = 29\,191.83$ km, $e = 0.2519$ for the internal. To change from one to another it is only necessary to apply the energy difference of $\Delta E = -0.02$. The sign means that there is a decrease of energy. In the three-body problem the osculating values will change and the trajectories are completely different, as can be seen in Fig. 4.

In the three-body problem the trajectory with two-body energy $E = -0.06$ will escape no matter we integrate it forward or backward, as we can see in Fig. 2. In Fig. 4 this trajectory was integrated with a negative time step. The other trajectory, with $E = -0.08$, integrated with a positive time step, is in the white region of Fig. 2. This trajectory will never escape (see Winter and Vieira Neto (2002)), and the spacecraft will become a satellite of the Moon. The collision trajectories also can be exploited. These trajectories can be used to deploy loads into the surface of the Moon.

The minimum possible energy for $\varpi = 0^\circ$ (Fig. 1) occurs at $R_p = 1\,838$ km (a perilune distance from 100 km of the surface of the Moon) with $E = -0.107$. Which orbital elements are $a \approx 21\,700$ km and $e \approx 0.915$, which is highly eccentric for an elliptic orbit. We have to note that most of the conditions for the minimum energy are long times trajectories in the vicinity of the Moon (more than 500 days).

For the case $\varpi = 180^\circ$ (Fig. 2) the minimum energy occurs at $R_p = 2\,738$ km with the value $E = -0.105$ canonical units (at a perilune distance of 900 km from the surface of the Moon). There is a collision region at $R_p = 1\,838$ km which prevents the decrease of the energy for this perilune. This trajectory has $a \approx 22\,100$ km and $e \approx 0.876$.

Other relevant information is shown in Figs. 5 and 6. From these figures it is possible to identify the

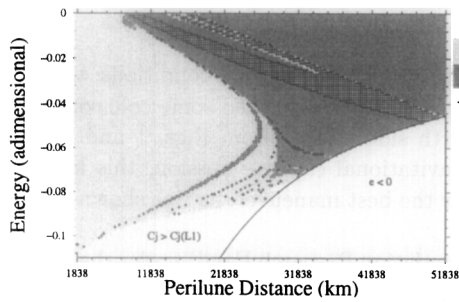


Fig. 5. Separation of the capture trajectories into two classes: captures from Earth side, and captures from anti-Earth side, for $\varpi = 0^\circ$.

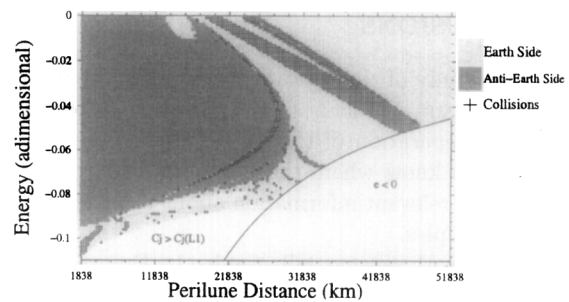


Fig. 6. Same as in Fig. 5, but for $\varpi = 180^\circ$.

capture direction, from the Earth side (through L_1), in light gray, or from the anti-Earth side (through L_2), in weight gray. The knowledge of this direction is very important, trajectories adopted by Belbruno and Miller use the anti-Earth side capture which provides good savings for the fuel consumption. The right choice of the perilune (in opposition or in inferior conjunction) can reach higher negative values for the two-body energy.

Some of the collision lines happen due to the separation of the two directions of capture. The Earth side captures reach the most negative energies for both cases. The anti-Earth side can reach energies much more negative for $\varpi = 180^\circ$.

MISSIONS EXAMPLES

Using the information given above it is possible to design various types of missions. Here we present some interesting possibilities. For example, suppose it is desired to place a spacecraft in a direct circular lunar orbit with altitude of 400 km ($R_p = 2138$ km) with pericenter at $\varpi = 0^\circ$. From Fig. 1 we have that the minimum possible energy to this mission is -0.106 , spending 45.04 days. From Fig. 5 we verify that this trajectory comes from the Earth-side.

Another possibility is to restrict the time in the sphere of influence. For example, suppose we want the capture above to have a time limit of five days. It can occur with energy -0.079 in a time of 4.11 days.

Other possibility is fixing the energy and finding the pericenter for a certain time limit. Example, suppose it is desired to make a maneuver with $E = -0.08$ in a time shorter than 10 days. In this case it is possible a whole range of perilune distances up to a maximum of 15 638 km in 9.74 days.

Finally, suppose a mission in the vicinity of the Moon, no matter where, but with a time duration greater than 500 days. In this case we can take a black point from Fig. 1 or 2 and choose the best for the mission purpose. For the period of time required these trajectories do not need any impulse to orbit around the Moon. The spacecraft is captured gravitationally, and after the mission time it will escape.

In the last mission case, we can see from Figs. 5 and 6 that the longest capture times occurs for the Earth side captures. Then, it is not possible to use the ballistic capture used by Belbruno and Miller to reach that region. But it is possible to use the ballistic captures described by Belbruno-Miller which ends near the trajectory we want, and then, instead of transfer directly to the Moon, we can make a transfer to the trajectory with a long capture time. This will save even more fuel consumption due to the proximity of the regions.

It is also possible to transfer directly to the white island. Trajectories in that region can stay indefinitely around the Moon without maintenance.

CONCLUSIONS

We study of the capture time for the Earth-Moon-Particle system for two perilune conditions, $\varpi = 0^\circ$ and $\varpi = 180^\circ$. We identified regions with different capture times. Figs. 5 and 6 related some collision lines with the separation of the capture type: Earth side and anti-Earth side cases. From Figs. 1 and 2 it is possible to know where the minimum energy will occur. In a gravitational capture mission, this kind of map offers relevant information about time, energy and location for the best maneuver, as was shown in the given examples.

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